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53B Remiger Road, Puhoi

Bush Quality Assessment Report

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1.0 Introduction

This report relates to three stands of native bush over 2ha in size on the property located at 53B Remiger Road and Tunnel Road, Puhoi. An additional stand smaller than 2ha is also proposed for protection under this application.

One of the stands is identified as Significant Ecological Area (SEA) under the Auckland Unitary Plan (Operative in Part) 2016 (AUP(OP)). The other stands are considered to meet sub-factors required to qualify as SEA, and also meet the significance criteria under the still operative Auckland Council District Plan (Rodney Section) 2013 (District Plan), as described in this report.

The protection of the bush areas described in this report will result in a valuable contribution to the preservation and improvement of native habitat, steep slopes, and watercourses in the surrounding landscape, where bush is largely fragmented and exotic pastureland and forestry dominate.

2.0 Overview of Bush on Property

The expected ecosystem type of this property prior to human disturbance is a mix of kauri, podocarp, broadleaved forest and taraire, tawa, podocarp forest[1]. The majority of this forest has been cleared for grazing in the past, and the majority of the remaining areas are currently grazed. Both ecosystem types are still present on the property, but are damaged by stock grazing, weed and pest invasion, and expected logging in the past. Only one stand is currently fenced from grazing.

The outer edges of the bush stands, as well as steeper slopes across the property, are currently regenerating with native vegetation with manuka, kanuka, totara, and lancewood being the most common canopy species in these areas. Regeneration is generally denser on those slopes less accessible to stock access.

A summary of the bush areas proposed for protection is included below:

Scheme Ref	Area	SEA?	Description
CU, CL, CM, CN, CO, CP, CQ, CR	37.23ha	Yes	Mature podocarp-broadleaf forest with a grazed understorey, with taraire forest on the lower slopes, covering the south facing slopes above the main stream flowing west to east across the property.
CT and CS	8.1ha	No	Mature podocarp-broadleaf forest with a grazed understorey, with regenerating kanuka forest around watercourses and on the outer edges.
CV	6.7ha	No	Mature podocarp-broadleaf forest with stock proof fencing and a dense regenerating understorey.
CW	1.4ha	No	Mature podocarp-broadleaf forest with a grazed understorey. Closely associated with Area CV.

[1] Singers, N.; Osborne, B.; Lovegrove, T.; Jamieson, A.; Boow, J.; Sawyer, J.; Hill, K.; Andrews, J.; Hill, S.; Webb, C. 2017. Indigenous.terrestrial and wetland ecosystems of Auckland. Published by Auckland Council.

An additional stand of kauri, podocarp, broadleaved forest is present in the north of the property, but is not proposed for protection under this application as it is beyond the area required to be protected for the five lots proposed.

The property is located in close proximity to existing private bush covenants, QEII reserves and Department of Conservation reserves. In the immediately surrounding landscape, however, plantation forest and pasture land dominate. This increases the importance of preserving bush in this location, as well as being located in the upper catchment for a significant tributary of the Puhoi River.

In terms of weed species, presence within the bush areas is relatively low due to frequent grazing. Pines (*Pinus radiata*) occur in the canopy of each area in low quantity, with a larger group in Area CV, and should be selectively poisoned.

The general approach to implementing the ACDP(RS) Appendix 7B Bush Quality Assessment is to get sample points that are representative of the variability that is commonly present throughout a forest patch. This allows for an overall average quality assessment under the assessment guidelines. In the case of this property, a separate bush assessment was carried out in each of the areas larger than 2ha proposed for covenanting. The AUP(OP) significance criteria are also discussed where relevant.

A description of each area followed by the assessment undertaken is shown below. The areas of bush referred to are as labelled on the accompanying Scheme Plan.

3.0 Bush Areas CU, CL, CM, CN, CO, CP, CQ and CR

3.1 Physical Description

These bush areas are part of a continuous stand of bush covering approximately 37.23ha across predominately south facing slopes with the peak of the hill being located in the northwest of the area. The main watercourse of the property runs from west to east through the southern part of the bush. To the north of the watercourse is an existing farm track that is to be fenced either side and excluded from the proposed covenant. For the most part, canopy closure is achieved across this track.

There is an existing fence around the eastern part of the bush, but stock are grazed within the bush and a track for stock movement runs from the eastern bite west and south to the south-western clearing which is not proposed for covenanting. The land to the north of this clearing appears to have recently naturally regenerated from pasture due to higher inaccessibility for stock.

3.2 Vegetation

The forest in these areas is primarily mature mixed podocarp and broadleaf forest. Stands of taraire forest are more common on the lower slopes in the south, and regenerating kanuka and totara are common on the edges.

The youngest vegetation occurs in the western central area, where regeneration across steep slopes appears to have occurred in a previously cleared area relatively recently. Pasture species and

regenerating gorse persist in this part of the bush, but other than this area weed presence is generally very low, excepting for the existing tracks. For the track running through the centre of the bush, this will be contained within the covenant and left to regenerate for which control of weed species such as kikuyu and foxglove will be essential. Other weed species observed include pine and Japanese honeysuckle.

Understorey cover within the bush area is generally low, due to the stock access, with only ground cover or unpalatable species occurring. The steeper parts of the bush, such as the northeast, where stock rarely venture, have a greater understorey community. More variation is generally found in the epiphytic and climbing species which have persisted in the bush despite disturbance to the understorey, and likely the canopy, in the past.

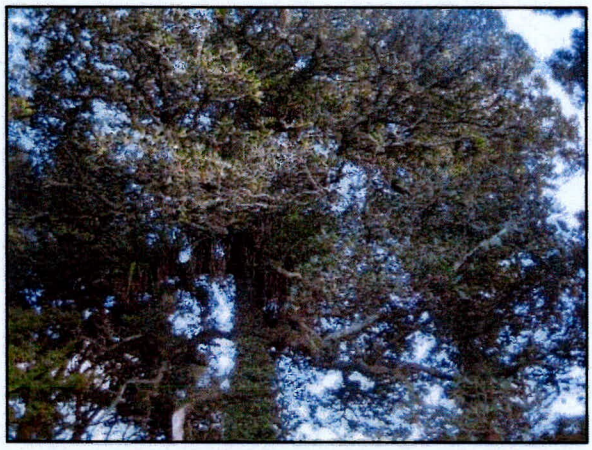
Canopy species observed in this bush area include totara, kanuka, karaka, puriri, taraire, silver fern, nikau, titoki, kohekohe, lancewood, rewarewa, maire, kahikatea, wheki, manuka, cabbage tree, rimu, rata, mapou, hangehange, and putaputaweta.

Common understorey, epiphytic and climbing species observed include rasp fern, thin-leaved coprosma, mapou, forest sedge, thin-leaved bastard grass, manuka, brake fern, basket grass, totara, hangehange, NZ gloxinia, hook grass, twiggy coprosma, scented fern, white rata, tank lily, hanging spleenwort, kiekie, leather leaf fern, and hound's tongue fern.

3.3 Photos

The following photos show examples of the canopy vegetation of the bush.









The following photos show examples of the understorey vegetation of the bush.









An overview of the bush looking north is shown below.



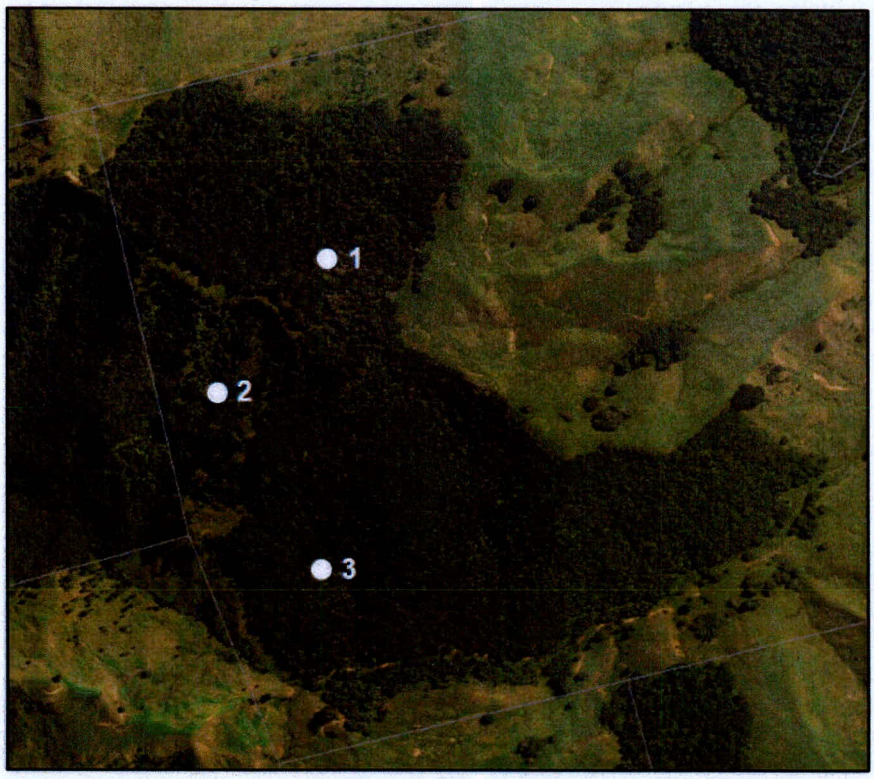
3.4 Significance Assessment

The entire area, stretching onto the western neighbouring property, is identified as SEA_ T_2535, which meets the SEA factor for Threat Status and Rarity, presumably because both kauri, podocarp, broadleaved forest and taraire, tawa, podocarp forest are endangered ecosystems in the Auckland Region[2] .

However, it is recognised that the SEA status of the bush area has no weighting in the ACDP (RS) bush quality assessment, which is still operative. This assessment is therefore outlined below.

3.4.1 Bush Quality Assessment Scale 1

The general approach to implementing the bush quality assessment is to get sample points that are representative of the variability that is commonly present throughout a forest patch. This allows for an overall average quality assessment under the assessment guidelines. Assessment at three canopy locations throughout the bush was undertaken as below.



[2] Singers, N.; Osborne, B.; Lovegrove, T.; Jamieson, A.; Boow, J.; Sawyer, J.; Hill, K.; Andrews, J.; Hill, S.; Webb, C. 2017. Indigenous terrestrial and wetland ecosystems of Auckland. Published by Auckland Council.

Bush Quality Assessment Field Sheet (Appendix 7B of District Plan Rodney, 2011)

SCALE 1

RANK	4	3	2	1	0	Site 1	Site 2	Site 3
CANOPY								
1. % of natives	81-100%	61-80%	41-60%	25-40%	<25%	4	4	4
2. % of canopy closure	81-100%	61-80%	41-60%	25-40%	<25%	4	2	3
3. Number of species	>12	8-11	4-7	2-3	1	3	2	2
4. Height	>8m	6.1-8m	4.1-6m	3.1-4m	<3m	4	2	4
UNDERSTOREY								
5. % of natives	81-100%	61-80%	41-60%	25-40%	<25%	4	2	4
6. % ground cover	81-100%	61-80%	41-60%	25-40%	<25%	2	1	0
7. Number of species	>12	8-11	4-7	2-3	1	4	3	2
					SITE TOTALS:	**25**	**16**	**19**

TOTAL A (average of all sites): 20

On scale 1, a final score of 20 is achieved. Based on this score, these areas exceed the quality standard required without the need to proceed to scale 2.

3.4.2 Bush Quality Assessment Scale 2

While significance of the bush is achieved under scale 1, it is also considered that five points would be awarded for the potential of regeneration in the parts of the bush that are in the early scrubby stages of succession (west) and throughout the understorey. This regeneration would be rapidly assisted through the completion of stock proof fencing around the bush. It is also considered five points could be awarded for aesthetic value due to the bush clad slopes of the peak located in the northwest of the area, and five points would be awarded for a continuous stand over 8ha.

3.4.3 Summary

Based on scale 1 and scale 2 scores, the total score for the bush area is 35. This bush area therefore meets the acceptance criteria for significance under the ACDP(RS) as well as being identified as SEA.

An ongoing management plan for the proposed bush covenant is provided in association with this application.

4.0 Bush Areas CS and CT

4.1 Physical Description

These bush areas cover approximately 8.1ha across east and west facing slopes around a central gully running north, along the western side of Area CS. The watercourse in this gully runs into the main watercourse of the property to the north. The two bush areas described in this section of the report are closely associated, being only separated by a farm track running through the centre, and are therefore considered continuous.

Neither of the bush areas are fenced and stock access is apparent in the general lack of understorey vegetation, at least around the edges of the bush stands. Narrow gullies and steep land within and around the edges of the bush have been colonised with regenerating native vegetation.

4.2 Vegetation

The forest in Areas CS and CT is primarily mature mixed podocarp and broadleaf forest. Regenerating kanuka forest with totara and lancewood featuring in the canopy is more common on the outer edges, particularly the north of Area CT and southeast of Area CS. Taraire forms the dominant canopy on lower elevation slopes, and kahikatea are common around the watercourse in the centre of the areas.

Canopy species observed in this bush area include kahikatea, nikau, rimu, taraire, totara, silver fern, rewarewa, kauri, puriri, manuka, kanuka, putaputaweta, lancewood, kohekohe, and wheki.

Understorey cover within the bush area is generally low, due to the stock access, with only ground cover or unpalatable species occurring. The steeper parts of the bush, such as the northeast, where stock rarely venture, have a greater understorey community. More variation is generally found in the epiphytic and climbing species which have persisted in the bush despite disturbance to the understorey, and likely the canopy, in the past.
Common understorey, epiphytic and climbing species observed include twiggy coprosma, white rata, forest sedge, ground ferns, manuka and kanuka, totara, basket grass, pigeonwood, meadow rice grass, hanging spleenwort, hound's tongue fern, thread fern, tank lily, kiekie, and *Griselinia.*

4.3 Photos

The following photos show examples of the canopy of Areas CS and CT.

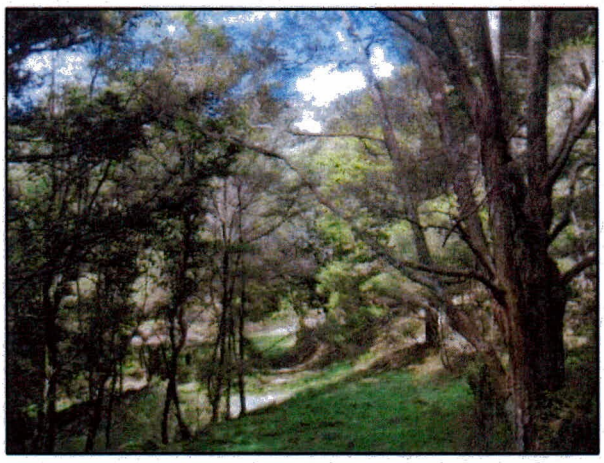








The following photos show examples of the internal areas and understorey of Bush Areas CS and CT.









4.4 Significance Assessment

This area is not identified as SEA so becomes non-complying under the AUP(OP). However, given it contains the same threatened ecosystem type as the larger area to the north, it is considered the area has merit under SEA sub-factor 2(a)[3]. The areas also provide a direct stepping stone between

[3] (2) **Threat Status and Rarity** (a) It is an indigenous habitat, community or ecosystem that occurs naturally in Auckland and has been assessed (using the IUCN threat classification system) to be threatened, based on evidence and expert advice (including Holdaway et al. Status assessment of NZ naturally uncommon ecosystems).

SEA_ T_2535 to the north and SEA_T_6779 and SEA_T_6773 to the south, so are considered to have merit under SEA sub-factor 4(c)[4]. The area's value as a stepping stone will increase upon covenanting which will ensure regeneration of the understorey vegetation.

The area has also been assessed under the still applicable ACDP(RS) criteria as below.

4.4.1 Bush Quality Assessment Scale 1

The general approach to implementing the bush quality assessment is to get sample points that are representative of the variability that is commonly present throughout a forest patch. This allows for an overall average quality assessment under the assessment guidelines. Assessment at three canopy locations throughout the bush was undertaken as below.



Bush Quality Assessment Field Sheet (Appendix 7B of District Plan Rodney, 2011)

SCALE 1

RANK	4	3	2	1	0	Site 1	Site 2	Site 3
CANOPY								
1. % of natives	81-100%	61-80%	41-60%	25-40%	<25%	4	4	4
2. % of canopy closure	81-100%	61-80%	41-60%	25-40%	<25%	3	3	4
3. Number of species	>12	8-11	4-7	2-3	1	3	2	3
4. Height	>8m	6.1-8m	4.1-6m	3.1-4m	<3m	4	4	4
UNDERSTOREY								
5. % of natives	81-100%	61-80%	41-60%	25-40%	<25%	2	2	4
6. % ground cover	81-100%	61-80%	41-60%	25-40%	<25%	0	0	0
7. Number of species	>12	8-11	4-7	2-3	1	2	2	3
					SITE TOTALS:	**18**	**17**	**22**

TOTAL A (average of all sites): 19

[4] (4) **Stepping-Stones, Migration Pathways and Buffers** (c) It is part of a network of sites that cumulatively provide important habitat for indigenous fauna or when aggregated make an important contribution to the provision of a particular ecosystem in the landscape.

On scale 1, a final score of 19 is achieved. Based on this score, Areas CS and CT do not exceed the quality standard required without the need to proceed to scale 2.

4.4.2 Bush Quality Assessment Scale 2

This bush area does not achieve the quality standards under Scale 1 only. It is considered five points would be awarded under scale 2 for the potential of rapid understorey regeneration once stock are excluded from the bush.

4.4.3 Summary

Based on scale 1 and scale 2 scores, the total score for the bush area is 24. This bush area therefore meets the acceptance criteria for significance under the ACDP(RS).

An ongoing management plan for the proposed bush covenant is provided in association with this application.

5.0 Bush Area CV

5.1 Physical Description

This bush area covers approximately 6.7ha in the east of the property across predominantly west facing slopes with a gully running south through the western part of the bush. A stream flows through this gully, entering the bush area from the southwest.

This area is entirely fenced from stock with a seven wire post and batten fence, plus an electric wire, allowing for dense regeneration of the understorey vegetation compared to the two bush stands previously described in this report.

5.2 Vegetation

The forest in Area CV is primarily mature mixed podocarp and broadleaf forest. Some larger trees occur in this area compared to those previously described, including several emergent rata and kauri trees. The central northern canopy contains a group of pines which are proposed to be poisoned.

Native canopy species observed in this bush area include taraire, nikau, rata, kahikatea, rimu, karaka, kauri, silver fern, cabbage tree, totara, mamaku, and pate.

Understorey cover is dense and includes species typical of this forest type such as nikau, mapou, mahoe, hangehange, and ground ferns and sedges.

5.3 Photos

The following photos show examples of the canopy of Area CV.







The following photos show examples of the internal areas and understorey of Area CV.





5.4 Significance Assessment

This area is not identified as SEA so becomes non-complying under the AUP(OP). However, given it contains the same threatened ecosystem type as the areas previously described, and in a better condition due to the stock proof fencing, it is considered the area has merit under SEA sub-factor 2(a)[5].

The area has also been assessed under the still applicable ACDP(RS) criteria as below.

5.4.1 Bush Quality Assessment Scale 1

The general approach to implementing the bush quality assessment is to get sample points that are representative of the variability that is commonly present throughout a forest patch. This allows for an overall average quality assessment under the assessment guidelines. Assessment at three canopy locations throughout the bush was undertaken as below.



Bush Quality Assessment Field Sheet (Appendix 7B of District Plan Rodney, 2011)

SCALE 1

RANK	4	3	2	1	0	Site 1	Site 2	Site 3
CANOPY								
1. % of natives	81-100%	61-80%	41-60%	25-40%	<25%	4	4	2
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3. Number of species	>12	8-11	4-7	2-3	1	2	2	2
4. Height	>8m	6.1-8m	4.1-6m	3.1-4m	<3m	4	4	4
UNDERSTOREY								

[5] (2) **Threat Status and Rarity** (a) It is an indigenous habitat, community or ecosystem that occurs naturally in Auckland and has been assessed (using the IUCN threat classification system) to be threatened, based on evidence and expert advice (including Holdaway et al. Status assessment of NZ naturally uncommon ecosystems).

5. % of natives	81-100%	61-80%	41-60%	25-40%	<25%	4	4	4
6. % ground cover	81-100%	61-80%	41-60%	25-40%	<25%	4	3	3
7. Number of species	>12	8-11	4-7	2-3	1	4	4	3
					SITE TOTALS:	**26**	**25**	**21**

TOTAL A (average of all sites): 24

On scale 1, a final score of 24 is achieved. Based on this score, Area CV exceeds the quality standard required without the need to proceed to scale 2.

5.4.2 Bush Quality Assessment Scale 2

This bush area achieves the quality standards under Scale 1 only. It is considered no further points would be awarded under scale 2.

5.4.3 Summary

Based on scale 1 and scale 2 scores, the total score for the bush area is 24. This bush area therefore meets the acceptance criteria for significance under the ACDP(RS).

An ongoing management plan for the proposed bush covenant is provided in association with this application.

6.0 Conclusion and Recommendations

Protecting the bush areas outlined in this report will have significant benefits for terrestrial fauna and flora and provide additional security for the conservation of bush in the wider landscape. The overall proposal will result in the protection of most of the significant bush on this property meaning weed and pest management will be more effective and the areas will provide beneficial ecological stepping stones across the entire site and connecting to surrounding SEAs and native vegetation.

Approximately 52ha of significant bush are present in the areas described in this report. These areas has vegetation quality exceeding the minimum requirements for acceptance based on the assessment criteria outlined in Appendix 7B of the District Plan and Schedule 3 of the Auckland Unitary Plan.

It is thus considered the bush areas proposed for covenanting meet the requirements for a bush protection subdivision under the operative plans.

This assessment is subject to the following recommendations:

I. Stock are excluded by permanent fence from all bush areas.
II. The covenant fence lines are regularly checked and maintained as stock proof.
III. Feral goats, deer, and other pest animals that may be present are controlled.
IV. Control of existing ecological weeds and pests occurs.
V. Regular monitoring for invasive weeds occur and any infestations are dealt with.
VI. Registration of bush protection covenants on the titles to the land.

7.0 Limitations

This report has been prepared for the particular project described to us and its extent is limited to the scope of work agreed between the client and Cato Bolam Consultants Limited.

No responsibility is accepted by Cato Bolam Consultants Limited or its directors, servants, agents, staff or employees for the accuracy of information provided by third parties and/or the use of any part of this report in any other context or for any other purposes.

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